

Micro Computer Architecture Assignment 03

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1 Introduction

The fast growth of intelligence, powerful computing and smart electronic systems is leading to new needs for computer designs and chip technology. Old ways of building computers are getting better with ideas that make processing faster use less power, access memory better and allow more changes in hardware. This report looks at five new technologies that are shaping how modern small computers and electronic systems are made. Neural Processing Units and Neural Engines offer parts to make artificial intelligence and machine-learning tasks go faster. Unified CPU, GPU and RAM setups help parts of a computer share data and talk to each other better. RISC-V gives an adjustable way to build instruction sets with support for special operations. Three-dimensional mixed integration and Through-Silicon Vias offer ways to connect chips, memory and other parts in a tight space. Finally materials that are not based on silicon, like gallium nitride, silicon carbide and flat materials are being studied to solve the limits of silicon technology.

2 Neural Processing Unit (NPU) and Neural Engines

2.1 What is an NPU?

Neural Processing Unit (NPU) is a processing unit like CPU and a GPU but specialized for Artificial Intelligence and Machine Learning Tasks. CPU designed to handle many different general purpose tasks and GPU mainly designed to perform graphics and parallel calculation when the NPU is mainly designed to perform mathematical operations used by neural networks such as, working with vectors and tensors. NPU is not a unit that replaces the CPU and the GPU; instead, they work together and each processor handles the type of work it is best suited for (heterogeneous computing). In modern devices such as smart phones , laptops and other smart electronic devices , the NPU is usually integrated with CPU and GPU inside a single System on Chip (Soc) to perform Artificial Intelligence tasks such as face recognition, speech recognition and image enhancement ect. Since Neural Processing Units (NPU) are specialized for AI workloads, they can provide high performance with relatively lower power consumption than CPUs and GPUs.

2.2 Architecture and Operation

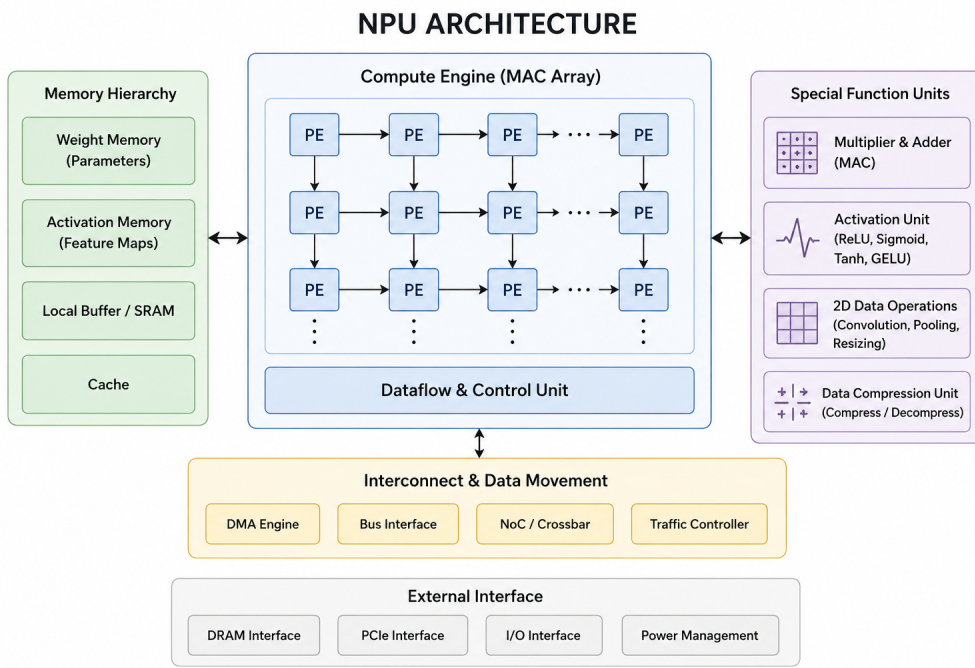


Figure 1: Simplified NPU Architecture

The Neural Processing Unit architecture is the internal organization of specialized hardware that is designed to perform neural network operations. A neural-network accelerator (NPU) contains a compute engine with many Processing Elements (PEs) that perform small calculations such as multiplications and accumulation (MAC), which are the fundamentals of neural-network calculations. Research on the Eyeriss accelerator demonstrates how parallel processing elements can be organized to efficiently execute convolutional neural networks while reducing energy consumption [1]. Similarly, Google’s TPU (Tensor Processing Unit) uses a large matrix of MAC units as its main computational engine. That shows the high importance of parallel computation in AI accelerators [2]. NPUs require an efficient memory hierarchy to store neural-network weights, input data, and intermediate results. Also, keeping frequently used data close to the processing unit reduces expensive data movements and improves energy efficiency [1]. In addition, the data flow and interconnect mechanism control how data moves between the memory and the processing elements. Eyeriss v2, for example, uses a flexible on-chip network to adapt data movement to different neural-network workloads [3]. Therefore, the NPU architecture mainly combines parallel processing elements, MAC units, local memory, data movement networks, and control mechanisms to achieve high performance and energy-efficient AI processing [1–3].

2.3 Applications

Neural Processing Units are mainly used to improve Artificial Intelligence and Machine Learning tasks. One important application of NPUs is Smartphones. Smartphones use NPUs for image enhancements, face recognition, speech processing, and other AI-based features with low power consumption [4]. NPUs are also useful in computer vision, including real-time image processing,

object detection and image classification [1]. In IoT and edge devices, they can use sensors and AI data locally by reducing latency and dependence on cloud servers [5]. NPUs can support healthcare, robotics, autonomous systems and smart environments, where fast AI interfaces are required [5]. As AI models become more common on modern edge devices, specialized neural-network accelerators are more important for achieving real-time performance within limited power and computing resources [5].

2.4 Advantages and Limitations

NPUs provide high performance and energy efficiency for AI work since they use specialized hardware and massive parallel processing for neural-network operations [6]. They can reduce data movement and reuse data locally, which helps to lower power consumption and improve processing efficiency [6]. NPUs are especially useful for real-time AI applications on mobile and edge devices where performance and battery life are important [7]. There are some limitations also. NPU may be less flexible than a CPU or GPU when running different or rapidly changing neural-network models [7], [8]. Because some accelerator designs work best only with specific model characteristics, such as particular data patterns or sparsity [8]. In addition, specialized hardware can require greater design complexity and become less efficient when the workload does not match its architecture [7]. Therefore, NPUs provide excellent AI performance per watt, but this advantage comes with trade-offs in flexibility, programmability, and workload compatibility.

2.5 Future Trends

In the future NPU development is expected to focus on higher energy efficiency, greater parallelism, and improved support for increasingly complex AI models [9]. NPUs are becoming more important in edge AI, allowing devices such as smartphones, IoT systems, robots and vehicles to process AI locally with low latency [10]. Researchers are exploring quantization, sparsity, model compression and memory-centric architecture to reduce the computation and data movement cost [9], [11]. Another emerging direction is on-device learning, which allows AI models to adapt directly to changing environments [12]. However, future NPUs face challenges including limited memory, high data movement cost, hardware complexity, model compatibility, scalability, and power constraints [9], [11]. As a result of that, future NPU research will increasingly focus on hardware-software co-design, flexible architectures, efficient memory systems and specialized support for emerging AI workloads.

3 Unified CPU, GPU and RAM Architectures

3.1 Traditional Architecture

In traditional computer architecture, the CPU and the GPU operate as separate processing units with different memory resources. The CPU normally access the system RAM, while the GPU has its own high-bandwidth memory. So the data that is required by the GPU must be transferred from CPU memory to the GPU memory through an interconnection such as PCI Express. After the GPU processing finishes, the processed data should be copied back to system

memory [1]. This separation provides high GPU bandwidth, but it introduces data-transfer overhead and increases programming complexity.

3.2 Unified Memory Architecture

Unified Memory Architecture (UMA) reduces this separation by allowing the CPU and the GPU to access a common memory. Instead of maintaining completely separate memory, both processors can operate on data located in shared system memory. Heterogeneous System Architecture (HSA) extends this concept by providing shared virtual memory addressing and mechanisms for cooperation between heterogeneous processing units [2]. A modern unified memory system can automatically manage memory placement and migration, reducing the need for programmers to explicitly copy data between CPU and GPU memories [3].

3.3 CPU-GPU Integration

CPU-GPU integration places the CPU cores and the GPU compute units within the same package (System-on-Chip). Integrated design reduces the physical communication distance and can provide a more tightly coupled processing environment. Heterogeneous System Architecture-based architectures allow different processing units to cooperate through shared memory and a common programming model [2]. Modern processors such as Apple silicon demonstrate this approach by combining CPU, GPU, and other accelerators around a unified memory architecture [4].

3.4 Performance Benefits

Unified architectures can improve performance by cutting down on data movement between different memory pools. When data doesn't have to move forth as much communication overhead goes down. This helps CPUs and GPUs work together better especially when the workload involves data exchanges. With shared memory both the CPU and GPU can use the physical memory space. That means memory is used efficiently since there's no need to duplicate data across separate systems. Also integrated designs often save energy because data doesn't always have to travel over an external connection [1] [2]. These benefits make unified architectures very useful for applications, in intelligence, graphics, multimedia processing and scientific computing.

3.5 Limitations

Unified architectures have some drawbacks even though they bring benefits. The CPU and GPU share the memory bandwidth. When both processors try to read or write amounts of data at the same time they can get blocked. That blockage is called contention. Also when a unified-memory system moves data from one place to another or when a page fault happens extra delay can appear. In some cases this delay can slow down the system. Moreover the shared memory pool may not give the GPU much fast high-bandwidth memory as a separate GPU that has its own dedicated VRAM. So while unified architectures make programming easier and help data flow between CPU and GPU the speed of the system depends a lot, on how the memory can move data on what the work looks like and on how the hardware was built.

4 RISC-V Vector Operations

4.1 Introduction to RISC-V

RISC-V is an Instruction Set Architecture. It is based on Reduced Instruction Set Computer principles. It was created at the University of California Berkeley. The goal was to create a modular and extensible instruction set. This instruction set is for computer architecture research. It is also for processor development. RISC-V is different, from ISAs. Commercial ISAs are controlled by companies. RISC-V is a standard. This allows organizations to design processors. These processors use the architecture. They do not need an ISA license. RISC-V is an open Instruction Set Architecture. RISC-V is an open Instruction Set Architecture. RISC-V is a standard. RISC-V is a standard.

4.2 Open ISA Concept

The open ISA idea takes the instruction-set rules. Keeps them separate from how the processor works. RISC-V creates a standard ISA and extra parts that can be added depending on what the program needs [2]. This means schools, companies that make computer chips and programmers can build different kinds of processors but still make sure the software works the same way on all of them. The RISC-V rules are out, in the open. Are made together by RISC-V International [3]. This openness helps people come up with ideas makes it easier to use different designs and lets people try out new ways to make processors.

4.3 Vector Extension

The RISC-V Vector Extension, also called the V extension adds instructions that help with data-parallel tasks. Of handling just one data item at a time these vector instructions can work on many items together using special registers known as vector registers [4]. This extension includes 32 vector registers. Allows different lengths for vectors and different sizes, for data elements. It works well for workloads that repeat the math operations over and over. Examples include computing, image processing, signal processing, machine learning and cryptography. The design keeps flexibility in mind so it can fit processors of sizes and implementations.

4.4 Customization

One of the benefits of RISC-V is that it can be customized. Normal extensions can be mixed with instructions that are made for a particular implementation using special encoding areas. For example a processor that is built for intelligence might have special instructions, for matrix or vector maths while a small embedded processor could have instructions that are tuned for control and sensor tasks. This kind of flexibility lets designers build processors that fit a field without having to change the whole software system.

4.5 Applications and Future

RISC-V can be used in embedded systems. It works well with microcontrollers too. It is also suitable for high-performance processors. RISC-V fits into accelerators. Supports edge

computing needs. It plays a role in intelligence applications. Researchers use it as a platform for experimentation. The standard vector architecture gives a base for improvements. These improvements include machine-learning tasks and cryptographic operations. As open-source hardware ecosystems grow RISC-V becomes more important. Specialized computing is expanding fast. RISC-V has potential. It could become a choice, for customizable systems. It also fits well with computing setups.

5 3D Heterogeneous Integration & TSVs

5.1 2D vs 2.5D vs 3D Integration

Traditional 2D integration puts transistors and functional parts on one flat silicon chip. This method works well. Is very old but as data moves between different parts it can slow things down and use more power. In 2.5D integration several chips sit next to each other on a base called an interposer. The interposer connects them closely without stacking the chips on top of each other. In integration the chips or layers are stacked up like a tower. Connections between the layers happen through paths made with technologies such, as Through-Silicon Vias (TSVs) [6] [7].

5.2 Heterogeneous Integration

Heterogeneous integration brings together components made using technologies, materials, process nodes or functions into a single system. For example, a package might combine processor logic, memory, sensors, radio-frequency parts, or accelerators that were produced using manufacturing processes [8]. This method supports the "More-than-Moore" idea, where system performance and functionality improve not by shrinking transistors but through smarter integration [9]. Chiplet-based architectures are an example of this since each chiplet can be designed and optimized separately before being assembled into a larger system.

5.3 Through-Silicon Vias

A Through-Silicon Via or TSV is an electrical connection that goes through a silicon die or wafer. These vias are usually filled with a material, most often copper and are used to link circuits or stacked dies in a vertical direction. TSVs allow for shorter communication paths compared to traditional package-level connections. The technology involves key steps like wafer thinning creating the via holes adding insulation depositing metal layers and bonding. Each of these steps needs to be controlled. Any defects in the TSVs can lead to problems, with performance and can reduce the long-term reliability of the device.

5.4 Performance Advantages

3D integration can greatly boost how many connections fit in an area while making the space between processing parts and memory parts much smaller. When connections are shorter data moves faster. Uses less energy. Stacking components vertically means more features can fit into a space [6]. High-Bandwidth Memory or HBM shows this idea in action. It uses memory chips

stacked on top of each other with vertical wires called TSVs. These allow data transfer between the memory and the processor or logic chip [10]. This kind of setup also lets different parts be built using the manufacturing methods available, for each component.

5.5 Challenges

Despite its benefits 3D integration brings a lot of problems. Getting rid of heat is harder because the dies that are stacked inside can stop heat from moving to the outside of the package. Thermal stress, bending, issues with through silicon vias, production checking for faults and getting enough good parts are also problems. These are mentioned in sources [8] and [9]. Also the steps needed to make the wafers thin stick them together. Create the through silicon vias can cost more money. So making 3D integration work well needs planning of the design the case, around the chip handling the heat making the parts and making sure they last a long time.

6 Beyond-Silicon Materials

6.1 Limitations of Silicon

Silicon has been the base of semiconductor technology because it has good electrical properties, reliable manufacturing methods and affordable wafer production.. As transistors keep getting smaller more physical and engineering problems appear. Short-channel effects, leakage currents, power density and problems, with controlling how transistors work become harder to deal with when the devices get very tiny [11]. Because of this scientists are looking for materials that can offer electrical, thermal, optical or mechanical qualities.

6.2 Wide-Bandgap Materials

Bandgap (WBG) semiconductors have much larger energy bandgaps than silicon. This feature helps devices handle electric fields and temperatures. It also allows them to switch faster and might lead to power losses [12]. Silicon Carbide (SiC) and Gallium Nitride (GaN) are two WBG materials. They are very useful, in power electronics, electric vehicles, renewable-energy systems, frequency electronics and power converters. Their material properties help designers create more efficient power systems compared with many conventional silicon-based designs [13].

6.3 2D Materials

Two-dimensional materials are made of layers that're extremely thin and can have special electrical and optical properties. Graphene, dichalcogenides such as MoS₂ and hexagonal boron nitride are examples of two-dimensional materials that scientists are studying for next-generation electronics [14]. Some two-dimensional semiconductors have bandgaps and can keep strong electrostatic control even when they are very small. Their thickness is one atom thick, which makes them appealing for future transistors and, for devices that stack on top of each other. However making these materials on a large scale dealing with contact resistance fixing material defects ensuring stability and fitting them into the usual semiconductor processes are still big challenges [14] [15].

6.4 Gallium Nitride / Silicon Carbide

GaN and SiC are now among the important practical alternatives to silicon for power semiconductor applications. I find that GaN devices are especially attractive for frequency and high-speed switching. SiC offers performance, in high-voltage and high-temperature environments [13]. I see that these technologies are used in electric vehicle power systems, fast chargers, renewable-energy converters, data-center power supplies and industrial power electronics. However reliability, manufacturing cost, defect control, packaging and device-specific limitations still require research.

6.5 Future Research

I think future semiconductor research will not rely on a material to replace silicon. Instead semiconductor scientists will mix materials based on what the device needs. WBG materials can help power conversion run smoothly while 2D materials can make small transistors, sensors, memory and allow heterogeneous integration. Researchers are also looking at ways to join emerging materials with CMOS technologies and advanced 3D architectures [15]. So the future of semiconductor technology will involve mixes of silicon, WBG materials, 2D materials and other emerging materials not a quick and total replacement of silicon.

7 Conclusion

The technologies discussed in this report show how microcomputer and semiconductor architectures are changing to meet the growing needs of computing systems. NPUs and Neural Engines give speed for artificial intelligence tasks while unified CPU, GPU and RAM designs cut data transfer time and help processing units work together better. RISC-V gives an open and flexible instruction set that lets users tailor it and use vector processing for different uses. Meanwhile 3D heterogeneous integration and TSV technology let processors, memory and other parts be combined with tighter connections although heat control and making the chips still remain big problems. Beyond-silicon materials, like GaN, SiC and two-dimensional materials give chances to improve power use, high-frequency operation and keep scaling devices. Together these developments show that future computing systems will rely more on processors, smart memory designs, open hardware, advanced packaging and new semiconductor materials instead of just making chips smaller.

References

- [1] Y.-H. Chen, “Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks,” *IEEE Journal of Solid-State Circuits*, 2017.
- [2] N. P. Jouppi, “In-datacenter performance analysis of a tensor processing unit,” *Proceedings of the 44th Annual International Symposium on Computer Architecture (ISCA)*, 2017.
- [3] J. E. Y.-H. Chen T.-J. Yang and V. Sze, “Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 2019.

- [4] Y.-H. C. et al, “Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 2019.
- [5] M. A. et al., “Embedded deep learning accelerators: A survey on recent advances,” *IEEE Transactions on Artificial Intelligence*, vol. vol. 5, no. no. 5, 2024.
- [6] J. S. E. Y.-H. Chen T. Krishna and V. Sze, “Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks,” *IEEE Journal of Solid-State Circuits*, vol. vol. 52, no. no. 1, Jan. 2017.
- [7] J. E. Y.-H. Chen T.-J. Yang and V. Sze, “Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. vol. 9, no. no. 2, 2019.
- [8] J. E. Y.-H. Chen T.-J. Yang and V. Sze, “Understanding the limitations of existing energy-efficient design approaches for deep neural networks,” in *Proc. SysML*, 2018.
- [9] T. Mohaidat and K. Khalil, “A survey on neural network hardware accelerators,” *IEEE Transactions on Artificial Intelligence*, vol. vol. 5, 2024.
- [10] T. M. et al, “Revisiting edge ai: Opportunities and challenges,” *IEEE Internet Computing*, vol. vol. 28, no. no. 4, 2024.
- [11] Y.-H. C. et al, “Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. vol. 9, no. no. 2, 2019.
- [12] J. Z. et al, “Dq-stp: An efficient sparse on-device training processor based on low-rank decomposition and quantization for dnn,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. vol. 71, no. no. 4, 2024.